

Vinh Nguyen

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EDUCATION

University of Texas at Austin

Master of Science in Artificial Intelligence • GPA: 3.93/4.0 • $\Phi K \Phi$ Honor Society

Austin, TX

Aug. 2024 – May 2026

Centennial College

Advanced Diploma in AI Software Eng Tech • GPA: 4.43/4.5 • CCSAI Scholarships

Toronto, ON

Aug. 2022 – Aug. 2024

Foreign Trade University

Bachelor in Economics

Ho Chi Minh, VN

EXPERIENCE

Machine Learning Student Researcher

Centennial College

Jul. 2024 – Aug. 2024

Toronto, ON

- Led a 7-week exploratory ML research project on the newly released dataset to analyze data quality, clarify the data availability with the data owner and evaluate opportunities for future SpO2 estimation development.

Data Analyst (Intern)

New Balance Canada

Jan. 2023 – Aug. 2024

Mississauga, ON

- Built automated reporting tools and dashboards using VBA, PowerQuery, PowerPivot, and Spotfire, converting a 3-hour manual reporting process into a 10-minute automated pipeline, enabling daily access to business reports.

Process and Technology Analyst

New Balance Vietnam

Apr. 2021 – Aug. 2022

Ho Chi Minh, VN

- Collaborated with global product and costing teams to analyze SOPs, optimize production workflows, and develop calculation tools, clearing a backlog of over 300 pending issues and earning the NB Associate Appreciation Award.

PROJECTS

Multi-Document RAG Assistant | Python, LangChain, ChromaDB, HF, Streamlit

[Demo](#) • [Code](#)

- Built a Retrieval-Augmented Generation (RAG) chatbot supporting pdf, docx, and txt document ingestion with semantic vector search and conversational QA.

LLM Tool-Calling Weather Assistant | Python, Groq, Geopy, Open-Meteo, Streamlit

[Demo](#) • [Code](#)

- Built an LLM-powered weather assistant with tool-calling capabilities using Groq API, Geopy geocoding, and Open-Meteo APIs, deployed through an interactive Streamlit interface.

Restaurant Review Sentiment Classification | PyTorch, HF Transformers, DVC, Gradio

[Demo](#) • [Code](#)

- Fine-tuned DistilBERT for 5-class sentiment classification on the Yelp Review Full dataset using Hugging Face Transformers and Trainer API.

Grade School Math Reasoning App | HF Transformers, LoRA, PEFT, DVC, Gradio

[Demo](#) • [Code](#)

- Fine-tuned small LLMs on GSM8K using LoRA-based supervised fine-tuning and evaluated instructional few-shot Chain-of-Thought (CoT) prompting, Direct Answer SFT, and CoT SFT reasoning strategies.

Scene Text Recognition Pipeline | PyTorch, YOLOv11, DVC

• [Code](#)

- Built a two-stage Scene Text Recognition pipeline by fine-tuning YOLOv11 for text detection and training a custom CRNN with CTC loss for text recognition on the ICDAR 2013 dataset.

Quantization and LoRA Techniques | Python, PyTorch, LoRA, Quantization

• [Code](#)

- Developed from-scratch implementations of FP16/BF16 half-precision training, 4-bit quantization, and LoRA adapters for parameter-efficient LLM fine-tuning.

TECHNICAL SKILLS

ML | AI: Python, PyTorch, Scikit-learn, HF Transformers, PEFT, LoRA, Quantization, Fine-tuning, RAG, LangChain, YOLO

Data Science: NumPy, Pandas, Matplotlib, MongoDB, ChromaDB, PowerBI, PowerQuery, Tableau, Spotfire

MLOps | Cloud: Docker, Git, MLflow, DVC, Hydra, FastAPI, AWS, Streamlit, Gradio